

Shell-Alignment from Projection Locking: A Discrete Admissibility Theorem under Born–Infeld Saturation

O22 — Spectral Admissibility Sub-programme

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Abstract

Within the spectral admissibility sub-programme of Cosmochrony, the canonical fibre-level observable $\sigma_{\text{pair}}^{\text{can}}(n)$ on the Cayley graph $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$ governs the charged-lepton mass hierarchy through a capacity decay exponent δ_{pair} . O21 conjectured that physical admissibility requires the observable saturation rank n_3^{obs} to align with a BFS shell of this graph — a condition left open as an independent geometric hypothesis.

We prove that this conjecture is not independent: it is a theorem. The proof identifies a *projection locking* mechanism intrinsic to the non-injective projection Π of the Cosmochrony framework. The argument rests on a distinction between two types of discreteness. The shell index n is discrete as a matter of graph geometry. The observable $\sigma_{\text{pair}}^{\text{can}}(n)$ is a discrete sample of a continuous effective decay law $\tilde{\sigma}(t) \sim C t^{-\delta_{\text{pair}}}$ inherited from the underlying χ -relaxation dynamics; the Born–Infeld admissibility constraint $A_n \leq c_\chi/\sqrt{\lambda_n}$ is therefore a condition on a continuously varying, generically irrational amplitude. Nothing in the Born–Infeld dynamics alone forces saturation to occur at an integer depth.

Shell-alignment follows from a third ingredient: the observable space of Π decomposes as $\mathcal{O} = \bigsqcup_{n \in \mathcal{N}_q} \mathcal{O}_n$, indexed by BFS shells, so that Π cannot encode a saturation event at a non-admissible depth. The physically realisable saturation depth is therefore the first element of $\mathcal{L}_{\text{BI}} \cap \mathcal{N}_q$, which lies in $\mathbb{N}$ by construction. At its core, the proof reduces to the incompatibility between a continuous saturation locus and a discrete admissibility support: the discreteness of n_{sat} is not assumed but derived as a constraint on admissible realisations under Π . As a corollary, the logarithmic commensurability condition of O21 is recovered as the observable signature of projection locking, not as an independent resonance principle. The entire derivation is foundational: the chain $c_\chi \rightarrow A_n^{\text{max}} \rightarrow \mathcal{L}_{\text{BI}} \cap \mathcal{N}_q \rightarrow n_{\text{sat}} \in \mathbb{N}$ introduces no parameter external to the Cosmochrony substrate. The problem of determining *which* shell is selected — equivalent to deriving $\Sigma_c(n_3) = 3$ — remains open and is the subject of O23.

Keywords. spectral admissibility; projection locking; Born–Infeld admissibility; non-injective projection; BFS shell structure; Heisenberg graphs; Weil representation; fibre-

level observable; continuous decay law; discrete realisation; shell-alignment; lepton mass hierarchy; cascade exponent; subdiffusive growth

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1 Introduction

1.1 The open problem after O21

The spectral admissibility sub-programme of Cosmochrony [1] has established, through O16–O21 [22, 23, 24, 25, 26, 27], that the physically relevant observable for the charged-lepton mass hierarchy is the canonical fibre-level pair observable $\sigma_{\text{pair}}^{\text{can}}(n)$, defined on conjugate Weil blocks $\{\rho_c, \rho_{q-c}\}$ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. The capacity exponent $\delta_{\text{pair}} \approx 7.44$, extracted from the power-law decay $\sigma_{\text{pair}}^{\text{can}}(n) \sim Cn^{-\delta_{\text{pair}}}$, lies within the phenomenological window [7.4, 10.6], and the structural relation $\beta^* = 1/(\delta_{\text{pair}} + 1/2)$ recovers the cascade exponent $\beta^* \approx 0.126$ consistent with the observed charged-lepton mass ratios.

O21 [27] constructed a fully observable and intrinsic saturation rank $n_3^{\text{obs}}(n_0, n_1, q)$, invariant under all pipeline normalisations, and stated the following conjecture:

Shell-Alignment Conjecture (O21 Conjecture 4.1). A conjugate pair $\{c, q - c\}$ is physically admissible if and only if $\text{dist}(n_3^{\text{obs}}, \{n \mid S_n \neq \emptyset\}) \ll 1$, where S_n denotes the BFS shell at depth n in $\text{Cay}(G_q, S_q)$.

Two problems were identified as open: establishing this conjecture, and deriving the threshold $\Sigma_c(n_3) = 3$ from the Born–Infeld admissibility structure. The present paper addresses the first problem.

1.2 Main result

We prove (Theorem 4.8) that every physically realisable saturation depth satisfies $n_{\text{sat}} \in \mathcal{N}_q \subset \mathbb{N}$, where $\mathcal{N}_q$ is the set of BFS shell depths of $\text{Cay}(G_q, S_q)$. The proof is an intersection argument: the Born–Infeld saturation locus $\mathcal{L}_{\text{BI}} \subset \mathbb{R}_{>0}$ is a priori real-valued, but physical realisability under Π constrains it to $\mathcal{N}_q$. Shell-alignment is therefore not a hypothesis about the spectral dynamics of ρ_c but a theorem about the interaction between the continuous Born–Infeld constraint and the discrete admissibility filter of Π .

It is important to note that this result is not trivial. The observable $\sigma_{\text{pair}}^{\text{can}}(n)$ is evaluated along a discrete sequence of shell depths $n \in \mathcal{N}_q$, but this geometric discreteness does not by itself imply that the saturation condition is satisfied at an integer depth. The Born–Infeld saturation locus $\mathcal{L}_{\text{BI}}$ is defined by the vanishing of a continuous amplitude function; its elements are generically irrational and carry no information about the shell structure of $\text{Cay}(G_q, S_q)$. The discreteness of n_{sat} is not assumed: it is derived as a constraint on admissible realisations under the projection Π , and it is this derivation that constitutes the content of the theorem.

1.3 Organisation

Section 2 recalls the relevant results of O16–O21 and fixes notation. Section 3 establishes the key distinction between the discrete shell index and the continuous effective decay law of $\sigma_{\text{pair}}^{\text{can}}$. Section 4 states and proves the Projection Locking theorem. Section 5 recovers the shell-alignment conjecture as a corollary and identifies the logarithmic commensurability of O21 as its observable signature. Section 6 discusses open directions, in particular the problem of shell selection (O23) and the large- q numerical programme.

2 Background

2.1 The Born–Infeld admissibility bound

The Cosmochrony framework [1, 2] derives the Born–Infeld effective action as the unique analytic completion compatible with bounded projective transport capacity. For a substrate field χ with field strength $F = D\chi$, the admissibility bound on mode amplitudes takes the form [3]

$$A_n \leq A_n^{\max} := \frac{c_\chi}{\sqrt{\lambda_n}}, \quad (1)$$

where λ_n is the Laplacian eigenvalue of the mode and c_χ is the saturation scale of the Born–Infeld action, encoding the maximal transport capacity of the substrate.

Remark 2.1 (Dynamic character of the bound). The bound (1) is *not* uniform in n : the admissible amplitude $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ varies with the spectral mode. As the BFS depth n increases and λ_n evolves with the shell geometry of $\text{Cay}(G_q, S_q)$, the admissibility envelope tightens or relaxes according to the spectral structure of the graph. This dynamic character is essential to the locking argument of Section 4.

2.2 The canonical fibre-level observable

The canonical pair observable $\sigma_{\text{pair}}^{\text{can}}(n)$ was introduced in O19 [25] as the pipeline-independent combination of the Gram–Schmidt spans of conjugate Weil blocks ρ_c and ρ_{q-c} on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. It is defined at the fibre level by eliminating the residual amplitude factor $r(c, q) \in \{1, 2, 3, 4\}$, which is a normalisation artefact of the O12/O13 pipeline [17, 18] rather than an arithmetic invariant of the Weil representation. By construction, $\sigma_{\text{pair}}^{\text{can}}(n)$ is:

- invariant under conjugation $c \leftrightarrow q - c$, reflecting the identity $\rho_{q-c} = \overline{\rho_c}$ established in O17 [23];
- independent of Gram–Schmidt seed and basis choices (O19 Proposition 7.2);
- evaluated at discrete shell depths $n \in \mathcal{N}_q$, while inheriting a continuous effective decay law from the underlying Weil representation dynamics (Section 3.2).

The distinction between the raw cumulative span $\Sigma^{(c)}(n) \in \mathbb{Z}$ and the normalised observable $\sigma_c(n) = \Sigma^{(c)}(n)/q$ is maintained throughout; $\sigma_{\text{pair}}^{\text{can}}(n)$ operates at the σ_c level.

2.3 The saturation rank and its continuous decay law

O21 [27] constructs the observable saturation rank $n_3^{\text{obs}}(n_0, n_1, q)$ by eliminating both the amplitude C and the exponent δ_{pair} from the power-law description of $\sigma_{\text{pair}}^{\text{can}}$. Given two pre-saturation measurements at depths $n_0 < n_1$ in the power-law regime, the rank is defined as

$$n_3^{\text{obs}}(n_0, n_1, q) := n_0 \left(\frac{q^2 \sigma_{\text{pair}}^{\text{can}}(n_0)}{9} \right)^{1/\delta(n_0, n_1)}, \quad (2)$$

where $\delta(n_0, n_1) = -\log(\sigma_{\text{pair}}^{\text{can}}(n_1)/\sigma_{\text{pair}}^{\text{can}}(n_0))/\log(n_1/n_0)$ is the local exponent estimated from the two measurements. By O21 Proposition 3.2, n_3^{obs} is independent of the choice of (n_0, n_1) within an exact power-law regime, and invariant under global rescaling of $\sigma_{\text{pair}}^{\text{can}}$.

The key structural point is that n_3^{obs} is derived from a *continuous* decay law: the local exponent $\delta(n_0, n_1)$ and the amplitude ratio are both functions of the continuously varying

effective envelope $\tilde{\sigma}$ of Lemma 4.5. Even though n_3^{obs} is computed from measurements at integer depths, it is in general a real number, and there is no reason internal to its definition for it to be an integer. Shell-alignment asserts that $n_3^{\text{obs}} \in \mathcal{N}_q$; this is the content of Corollary 5.1.

3 Two Types of Discreteness

The argument of Section 4 depends on a distinction that must be made precise.

3.1 Geometric discreteness: the shell index

The Cayley graph $\text{Cay}(G_q, S_q)$ with $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ carries a natural BFS shell structure $S_0, S_1, S_2, \dots$ determined by the word metric d on G_q . The shell index $n \in \mathbb{N}$ is discrete as a matter of graph geometry: there are no vertices at non-integer depths. The observable $\sigma_{\text{pair}}^{\text{can}}(n)$ is therefore evaluated along a discrete sequence of depths.

3.2 Dynamical continuity: the effective decay law

The power-law decay $\sigma_{\text{pair}}^{\text{can}}(n) \sim Cn^{-\delta_{\text{pair}}}$ is a consequence of the Weil representation dynamics of ρ_c on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ [22, 23]. This decay law is *continuous* in the effective sense: it interpolates smoothly between consecutive shell depths, and its parameters $(C, \delta_{\text{pair}})$ are determined by the continuous spectral structure of the representation, not by the discrete shell boundaries.

Remark 3.1 (Non-triviality of shell-alignment). The fact that n is geometrically discrete does not make shell-alignment trivial. The Born–Infeld saturation condition $A_n \approx A_n^{\text{max}}$ is a condition on a continuously varying amplitude. Nothing in the decay law of $\sigma_{\text{pair}}^{\text{can}}$ forces this condition to be satisfied exactly at an integer depth rather than strictly between two consecutive shells. Shell-alignment asserts that the continuous saturation condition and the discrete shell geometry are commensurate at the saturation point. This commensurability requires an explanation; providing it is the content of Theorem 4.8.

4 Projection Locking

4.1 Admissible realisation under Π

The central concept of this section is that of a *physically admissible realisation* of a saturation event. Saturation is not merely the mathematical condition $A_n = A_n^{\text{max}}$; it is an event that must be *encoded* by the projection Π as a distinguishable effective state. We formalise this as follows.

Definition 4.1 (Projectively admissible depth). *A depth $n \in \mathbb{R}_{>0}$ is projectively admissible if the BFS shell $S_n = \{g \in G_q \mid d(e, g) = n\}$ is non-empty, i.e. if $n \in \mathbb{N}$ and $S_n \neq \emptyset$ in $\text{Cay}(G_q, S_q)$. The set of projectively admissible depths is denoted*

$$\mathcal{N}_q := \{n \in \mathbb{N} \mid S_n \neq \emptyset\} \subset \mathbb{N}. \quad (3)$$

Definition 4.2 (Admissible realisation of a saturation event). *A saturation event at depth n_{sat} is physically realisable if there exists a substrate configuration $\chi \in \Omega$ such that:*

(R1) the Born–Infeld bound is reached: $A_{n_{\text{sat}}} = c_\chi / \sqrt{\lambda_{n_{\text{sat}}}}$;

(R2) the projection $\Pi(\chi)$ encodes a non-trivial effective state supported on the shell $S_{n_{\text{sat}}}$, i.e. $\Pi(\chi) \in \mathcal{O}_{n_{\text{sat}}}$ where $\mathcal{O}_{n_{\text{sat}}}$ is the component of the observable space $\mathcal{O}$ indexed by $S_{n_{\text{sat}}}$.

Remark 4.3 (Observable space decomposition). The observable space $\mathcal{O}$ of the Cosmochrony framework decomposes as

$$\mathcal{O} = \bigsqcup_{n \in \mathcal{N}_q} \mathcal{O}_n, \quad (4)$$

where each component $\mathcal{O}_n$ encodes the effective states resolvable at shell depth n . This decomposition is a consequence of the non-injectivity of Π : since Π collapses the fibre $\Pi^{-1}(o)$ to a single effective state $o \in \mathcal{O}$, and since the Gram–Schmidt span tracker measures novelty *per BFS shell*, distinct shells index distinct components of $\mathcal{O}$. Crucially, there is no component $\mathcal{O}_{n'}$ for $n' \notin \mathcal{N}_q$: the projection Π cannot produce an effective state at a non-admissible depth because no substrate configuration has a BFS-distance to the identity that is a non-integer real number.

4.2 Three preparatory lemmas

Lemma 4.4 (Discrete support of Π). *For any substrate configuration $\chi \in \Omega$, the effective state $\Pi(\chi)$ is supported on $\mathcal{N}_q$:*

$$\text{supp}(\Pi(\chi)) \subset \mathcal{N}_q \subset \mathbb{N}. \quad (5)$$

In particular, Π cannot encode an effective state at any depth $n' \notin \mathcal{N}_q$.

Proof. The Gram–Schmidt span tracker of the O12/O13 pipeline [17, 18] measures the rank of the projection of the Weil representation ρ_c onto the subspace spanned by group elements at BFS distance $\leq n$ from the identity. This rank is an integer-valued function of $n \in \mathcal{N}_q$, defined only at shell boundaries. The canonical observable $\sigma_{\text{pair}}^{\text{can}}(n)$ is constructed from this rank via the normalisation of O19 [25], and is therefore defined only at $n \in \mathcal{N}_q$. No effective state of the form $\Pi(\chi) \in \mathcal{O}_{n'}$ exists for $n' \notin \mathcal{N}_q$, since the corresponding component $\mathcal{O}_{n'}$ is empty by Definition 4.1. $\square$

Lemma 4.5 (Continuous effective decay law). *There exists a continuous, monotone decreasing function $\tilde{\sigma} : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that*

$$\tilde{\sigma}(n) = \sigma_{\text{pair}}^{\text{can}}(n) \quad \text{for all } n \in \mathcal{N}_q, \quad (6)$$

and such that in the pre-saturation regime

$$\tilde{\sigma}(t) = C t^{-\delta_{\text{pair}}} (1 + O(t^{-1})), \quad t \rightarrow \infty, \quad (7)$$

with $C > 0$ and $\delta_{\text{pair}} > 0$. In particular, the amplitude $A(t)$ associated with $\tilde{\sigma}(t)$ varies continuously in t and is generically irrational; the saturation condition $A(t) = c_\chi / \sqrt{\lambda(t)}$ is a condition on a real-valued, not integer-valued, parameter.

Sketch. The observable $\sigma_{\text{pair}}^{\text{can}}(n)$ is constructed from averaged spectral quantities associated with the χ -relaxation dynamics of the Cosmochrony substrate [1]. This dynamics is intrinsically continuous in the relaxation parameter τ ; the discrete index $n \in \mathcal{N}_q$ arises from the sampling induced by the projection Π onto BFS shells, not from a discreteness in the underlying dynamics. The values $\sigma_{\text{pair}}^{\text{can}}(n)$ therefore constitute a discrete sample of a continuously defined process, and $\tilde{\sigma}$ is an interpolating envelope of this sample. The construction does not rely on any specific representation-theoretic realisation; it requires only the existence of a continuous relaxation parameter and the monotone decay of admissible amplitudes under the Born–Infeld bound (1). Monotonicity of $\tilde{\sigma}$ follows directly from the latter: admissible amplitudes decrease monotonically as the saturation bound tightens with increasing spectral eigenvalue along the BFS trajectory. $\square$

Remark 4.6 (Independence from representation-theoretic smoothness). The existence of $\tilde{\sigma}$ does not depend on the smoothness of Weil representation matrix coefficients, nor on any specific algebraic property of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. It is a consequence of the continuity of the underlying χ -relaxation flow, which is a foundational assumption of the Cosmochrony framework [1] and is independent of the discrete group structure used to model it.

Lemma 4.7 (Born–Infeld saturation is a real-valued condition). *The Born–Infeld saturation locus*

$$\mathcal{L}_{\text{BI}} := \{t \in \mathbb{R}_{>0} \mid A(t) = c_\chi / \sqrt{\lambda(t)}\} \quad (8)$$

is generically a discrete subset of $\mathbb{R}_{>0}$, but contains no information about the shell structure of $\text{Cay}(G_q, S_q)$. In particular, the elements of $\mathcal{L}_{\text{BI}}$ are not constrained to $\mathcal{N}_q$ by the Born–Infeld dynamics alone.

Proof. The saturation condition $A(t) = c_\chi / \sqrt{\lambda(t)}$ equates two independently defined functions of t : the effective amplitude $A(t)$ derived from $\tilde{\sigma}(t)$ via Lemma 4.5, and the spectral bound $c_\chi / \sqrt{\lambda(t)}$ derived from the Laplacian eigenvalue profile of $\text{Cay}(G_q, S_q)$. Both functions are continuous in t . Their intersection $\mathcal{L}_{\text{BI}}$ is generically a finite set of real numbers, determined by the dynamics of $\tilde{\sigma}$ and λ , with no reason to coincide with $\mathcal{N}_q$ unless an additional constraint enforces it. That additional constraint is provided by Π , as established in Theorem 4.8 below. $\square$

4.3 Main theorem

Theorem 4.8 (Projection Locking under Born–Infeld Spectral Admissibility). *Let $\sigma_{\text{pair}}^{\text{can}}(n)$ be the canonical fibre-level observable on $\text{Cay}(G_q, S_q)$, subject to the Born–Infeld admissibility constraint (1) for each spectral mode, and let $\tilde{\sigma}$ be its continuous effective envelope (Lemma 4.5). Let Π be the non-injective projection of Cosmochrony, whose admissible outputs decompose as in (4).*

Then every physically realisable saturation depth satisfies

$$n_{\text{sat}} \in \mathcal{N}_q \subset \mathbb{N}. \quad (9)$$

That is, the Born–Infeld saturation condition can be physically realised only at depths at which a BFS shell of $\text{Cay}(G_q, S_q)$ is available to support the projection.

Proof. Let $t^* \in \mathcal{L}_{\text{BI}}$ be a point in the saturation locus (Lemma 4.7). We show that if the saturation event at t^* is physically realisable in the sense of Definition 4.2, then $t^* \in \mathcal{N}_q$.

By condition (R2) of Definition 4.2, physical realisability requires $\Pi(\chi) \in \mathcal{O}_{t^*}$. By Remark 4.3, the component $\mathcal{O}_{t^*}$ is non-empty if and only if $t^* \in \mathcal{N}_q$. If $t^* \notin \mathcal{N}_q$, then $\mathcal{O}_{t^*} = \emptyset$, and condition (R2) cannot be satisfied by any $\chi \in \Omega$: there is no effective state available to encode the saturation event.

Therefore, a saturation event at t^* is physically realisable only if $t^* \in \mathcal{N}_q$. Since the physically realisable saturation depth is by definition the first $t^* \in \mathcal{L}_{\text{BI}} \cap \mathcal{N}_q$, we conclude $n_{\text{sat}} \in \mathcal{N}_q \subset \mathbb{N}$. $\square$

Remark 4.9 (Saturation versus mathematical zero of a function). The locking theorem distinguishes between two notions that must not be conflated. The Born–Infeld saturation locus $\mathcal{L}_{\text{BI}}$ is a mathematical set of real numbers at which a continuous amplitude condition is satisfied; it is well-defined for all $t \in \mathbb{R}_{>0}$. Physical realisation, by contrast, requires the saturation event to be *encoded* by Π as a distinguishable effective state. This encoding is available only at $\mathcal{N}_q$. The locking is therefore not a property of the decay law alone, nor of the shell structure alone, but of their interaction through Π .

Remark 4.10 (Independence from $\sigma_{\min}$). The proof uses only the *existence* of a saturation locus $\mathcal{L}_{\text{BI}} \neq \emptyset$, which follows from the Born–Infeld bound (1) and the decay of $\tilde{\sigma}$. It does not require knowing the value of $\sigma_{\min} := \tilde{\sigma}(n_{\text{sat}})$, nor its expression in terms of c_χ and $\lambda_{n_{\text{sat}}}$. The derivation of this explicit value is a separate problem, deferred to a subsequent analysis.

Remark 4.11 (Role of c_χ). The saturation scale c_χ is not a free parameter in the Cosmochrony framework: it is determined by the maximal transport capacity and connectivity density of the underlying relational substrate [1, 2]. The foundational chain is therefore

$$c_\chi \longrightarrow A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}} \longrightarrow \mathcal{L}_{\text{BI}} \subset \mathbb{R}_{>0} \longrightarrow \mathcal{L}_{\text{BI}} \cap \mathcal{N}_q \longrightarrow n_{\text{sat}} \in \mathbb{N}, \quad (10)$$

and no parameter external to the Cosmochrony substrate is introduced at any step.

5 Shell-Alignment as a Corollary

Corollary 5.1 (Shell-alignment conjecture resolved). *Under the hypotheses of Theorem 4.8, the observable saturation rank $n_3^{\text{obs}}(n_0, n_1, q)$ of O21 Definition 3.1 [27] satisfies*

$$\text{dist}(n_3^{\text{obs}}, \{n \mid S_n \neq \emptyset\}) \ll 1, \quad (11)$$

i.e. shell-alignment holds as a consequence of projection locking.

Proof. By Theorem 4.8, $n_{\text{sat}} \in \mathcal{N}_q$. By O21 Proposition 3.2 [27], $n_3^{\text{obs}} = n_{\text{sat}}$ whenever $\sigma_{\text{pair}}^{\text{can}}$ follows an exact power law on the pre-saturation window (n_0, n_1) . In this regime the distance to the nearest BFS shell is therefore zero: $\text{dist}(n_3^{\text{obs}}, \mathcal{N}_q) = 0$. $\square$

Remark 5.2 (Finite-size effects and the residual at $q = 61$). The numerical value $\text{dist}(n_3^{\text{obs}}, \mathbb{Z}) \approx 0.20$ observed at $q = 61$ [27] is consistent with the exact-power-law hypothesis of O21 Proposition 3.2 being satisfied only approximately at small primes. Two sources of finite-size deviation are expected: the pre-saturation window (n_0, n_1) is short (of order $O(\log q)$ BFS steps, as established in O8 [13]), and the discrete shell spacing is coarse relative to the continuous saturation locus $\mathcal{L}_{\text{BI}}$. Both effects are expected to decrease as $q \rightarrow \infty$. The systematic verification of this convergence is part of the large- q numerical programme described in Section 6.2; it constitutes the primary empirical test of Corollary 5.1.

Remark 5.3 (Logarithmic commensurability recovered). O21 Remark 4.2 [27] observed that shell-alignment is equivalent to a logarithmic commensurability condition: the local decay rate δ_{pair} and the amplitude ratio $A = q^2\sigma(n_0)/9$ must be commensurate with the integer shell structure. Corollary 5.1 shows that this commensurability is not an independent resonance condition imposed from outside, but the *observable signature* of projection locking: the continuous decay law and the discrete shell geometry are commensurate at the saturation point *because* the locking mechanism forces the saturation event onto a shell. The terminology “resonance” in O21 is therefore a description at the level of the observable; the present paper provides the underlying mechanism.

6 Open Directions

6.1 Shell selection: the problem of O23

Theorem 4.8 establishes that $n_{\text{sat}} \in \mathbb{N}$, i.e. that saturation occurs *on* a BFS shell. It does not determine *which* shell. Determining the selected shell requires identifying the depth n_{sat} as a function of $(c_\chi, \lambda_n, \text{graph geometry})$, and in particular explaining why $\Sigma_c(n_3) = 3$ — the threshold underlying the definition of n_3^{obs} in O21. This is the subject of O23: once the locking mechanism is established, the question “why a shell?” is resolved and the question “why *this* shell?” becomes well-posed. The number 3 is interpreted in O21 as the number of physically stable directions selected by Π in the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$; O23 will determine whether this is a derived consequence of the Born–Infeld structure or an independent arithmetic property of the Weil representation.

6.2 Large- q numerical programme

The numerical evidence for shell-alignment currently rests on a single pair $(c, q - c) = (1, 60)$ at $q = 61$, with $\text{dist}(n_3^{\text{obs}}, \mathbb{Z}) \approx 0.20$. This intermediate value neither confirms exact alignment nor refutes it: it is consistent with a finite-size deviation from the exact locking point, expected to decrease as $q \rightarrow \infty$. An extended numerical campaign at larger primes and across multiple conjugate pairs is required to provide empirical confirmation of Corollary 5.1 and to constrain the rate of convergence. This programme is a prerequisite for the full quantitative validation of O23.

6.3 Generalisation beyond $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$

The locking argument of Section 4 uses the structure of Π as a discrete admissibility filter, but does not depend on the specific group structure of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. A generalisation to other groups carrying a Weil-type representation and a natural BFS shell structure is therefore in principle accessible, and would strengthen the universality of the projection locking mechanism.

7 Conclusion

This paper establishes that the shell-alignment conjecture of O21 [27] is not an independent geometric condition but a structural consequence of projection locking under Born–Infeld spectral admissibility.

The argument proceeds in three steps. First, we distinguish two types of discreteness: the shell index $n \in \mathcal{N}_q$, which is geometric, and the effective decay law of $\sigma_{\text{pair}}^{\text{can}}$, which is continuous (Lemma 4.5). The Born–Infeld saturation condition $A_n \approx c_\chi / \sqrt{\lambda_n}$ is a condition on the continuously varying amplitude, and the saturation locus $\mathcal{L}_{\text{BI}} \subset \mathbb{R}_{>0}$ carries no information about the discrete shell structure (Lemma 4.7). Second, we formalise the non-injective projection Π as a discrete admissibility filter: its observable space decomposes as $\mathcal{O} = \bigsqcup_{n \in \mathcal{N}_q} \mathcal{O}_n$, and no effective state exists at non-admissible depths (Lemma 4.4, Remark 4.3). Third, we combine these two structures: a saturation event is physically realisable only if condition (R1) (Born–Infeld bound reached) and condition (R2) (projection support available) are simultaneously satisfied, which forces $n_{\text{sat}} \in \mathcal{N}_q$ (Theorem 4.8).

Shell-alignment is therefore not imposed by the spectral dynamics of the Weil representation alone, nor by the geometry of $\text{Cay}(G_q, S_q)$ alone, but emerges as a necessary consequence of their interaction through Π under the Born–Infeld constraint. The “resonance” language of O21 Remark 4.2 is recovered as the observable signature of this locking at the level of n_3^{obs} (Corollary 5.1, Remark 5.3): what appears as commensurability between a continuous decay rate and a discrete shell structure is the projection of a single mechanism onto two different descriptive levels.

The locking mechanism is entirely foundational in the sense of Remark 4.11: the chain $c_\chi \rightarrow \mathcal{L}_{\text{BI}} \rightarrow \mathcal{L}_{\text{BI}} \cap \mathcal{N}_q \rightarrow n_{\text{sat}} \in \mathbb{N}$ introduces no parameter external to the Cosmochrony substrate.

Two problems remain open after this paper. The first is the determination of *which* shell is selected: identifying n_{sat} as a function of $(c_\chi, \lambda_n, \text{graph geometry})$, and in particular deriving the threshold $\Sigma_c(n_3) = 3$ from the Born–Infeld admissibility structure. This is the subject of O23. The second is the empirical confirmation of Corollary 5.1 via an extended numerical campaign at larger primes, which will quantify the rate of convergence of $\text{dist}(n_3^{\text{obs}}, \mathcal{N}_q)$ to zero and serve as the primary falsifiability test of the present result.

Within the spectral admissibility sub-programme, the present paper closes the first of the two open problems stated in O21: the shell-alignment conjecture is now a theorem. The programme enters its foundational phase, in which the remaining open problem — deriving the selected shell depth from first principles — is sharply delimited and well-posed.

References

- [1] J. Beau, “Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics,” white paper, Zenodo (2025). DOI: 10.5281/zenodo.17957509.
- [2] J. Beau, “Emergent Spacetime Geometry from Bounded Relational Relaxation,” companion paper, Zenodo (2026). DOI: 10.5281/zenodo.18407505.
- [3] J. Beau, “Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of $\text{SU}(2)$ Subgroups,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18944985.
- [4] J. Beau, “Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18945273.
- [5] J. Beau, “Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19040490.

- [6] J. Beau, “Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19056975.
- [7] J. Beau, “Projective Resolution Dynamics on Variable-Valence Relaxation Graphs,” O1 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076319.
- [8] J. Beau, “Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence,” O3 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076244.
- [9] J. Beau, “Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β ,” O4 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19101472.
- [10] J. Beau, “Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement,” O5 spectr. adm. subprg. paper, Zenodo (2026). DOI: 0.5281/zenodo.19120328.
- [11] J. Beau, “Matrix-Level Dynamic Redundancy and the Structural Confinement of the Cascade Exponent β ,” O6 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19140624.
- [12] J. Beau, “Projective Capacity and the Continuum Limit of Admissible Redundancy,” O7 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19149773.
- [13] J. Beau, “Three-Step Path Fingerprints on LPS Graphs and the Capacity Growth Exponent,” O8 spectr. adm. subprg. paper, Zenodo (2025). DOI: 10.5281/zenodo.19151769.
- [14] J. Beau, “Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent,” O9 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19154517
- [15] J. Beau, “Polynomial-Growth Capacity Dynamics at Large q : Volumetric Growth Confirmation, Capacity Decay, and the Weil Decomposition Roadmap,” O10 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19161074
- [16] J. Beau, “Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Representation Obstruction and Extracting the Capacity Exponent,” O11 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19169987
- [17] J. Beau, “Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction,” O12 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19194798.
- [18] J. Beau, “Asymptotic Stability of Exact Weil-Block Capacity on Heisenberg Graphs: Extended Prime Range, Variance Reduction, and Requalification of the $\delta \rightarrow \beta^*$ Tension,” O13 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19210831.
- [19] J. Beau, “Observable-Class Mismatch and the Corrected $\delta \mapsto \beta^*$ Relation on Heisenberg Graphs: Block Normalisation, Central-Phase Contribution, and Structural Correspondence,” O14 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19226272.

- [20] J. Beau, “Topological Invariants of Admissible Configurations: Charge Quantisation, Absence of Magnetic Monopoles, and Projective Chirality in Cosmochrony,” *compagnon paper*, Zenodo (2026). DOI: 10.5281/zenodo.19282962.
- [21] J. Beau, “Observable-Class Derivation Mismatch and the Exact Growth Equation: From Proxy Capacity to Block-Level Dynamics,” *O15 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19281420.
- [22] J. Beau, “Fibre Admissibility and Resolution of the Capacity Exponent Discrepancy,” *O16 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19298427.
- [23] J. Beau, “Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable,” *O16 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19299611
- [24] J. Beau, “Minimal Fibre Structure of the Non-Injective Projection from Born–Infeld Indiscernability: Derivation of the Parity Involution,” *O18 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19321413
- [25] J. Beau, “About Canonical Pair Observables in Weil Blocks: Structure of the Residual Amplitude Factor and Pipeline-Independent Formulation,” *O19 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19323720
- [26] J. Beau, “Projective Persistence and the Physical Sub-Spectrum: A Dynamical Selection Criterion for the Capacity Exponent,” *O20 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19339888
- [27] J. Beau, “Shell-Level Saturation and the Fibre-Level Transfer $\delta_{\text{pair}} \rightarrow \beta^*$,” *O20 spectr. adm. subprg. paper*, Zenodo (2026). DOI: 10.5281/zenodo.19358961